DPVSA

StaticBAC V2

Optimized binary arithmetic coding and entropy-aware quantization for neural network compression



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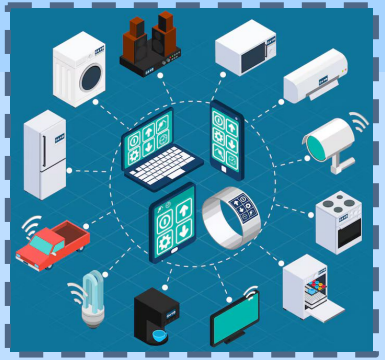
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Introduction



Motivation

- **Neural networks (NNs)** are increasingly deployed **on Edge devices**
  - Memory, energy and latency constraints
- Modern DNNs and LLMs often contain 100's of millions to **trillions of parameters**
  - **Memory access is the bottleneck**: model storage and memory traffic are barriers for efficient deployment

Need for Efficient Compression of NN parameters

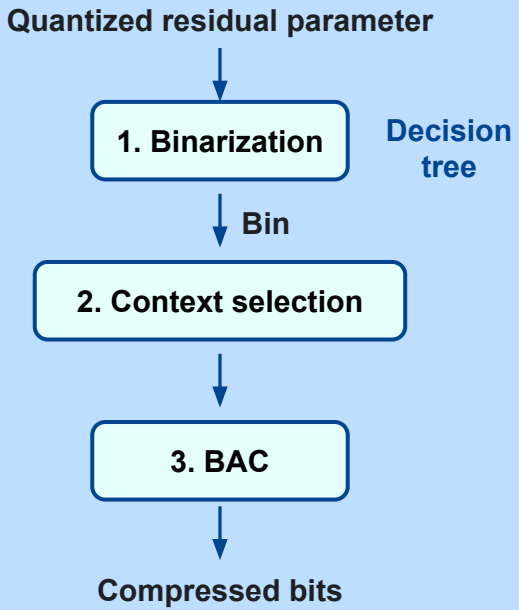
- Reduce model storage requirements and memory traffic
- Enable efficient execution on resource-constrained hardware

Context

StaticBAC - Basic concepts

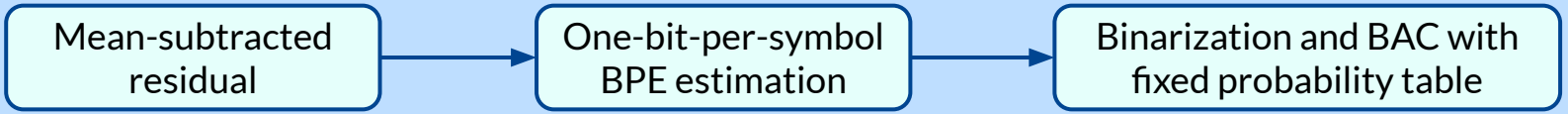
- Uses binary **arithmetic coding** (AC) for the compression of NN parameters.
  - **Entropy coding: lossless** scheme based on statistical properties of the data.
- Based on **DeepCABAC:** NNCodec's AC core for compression of NN parameters. ISO/IEC 15938-17 (MPEG-7 Part 17)
  - CABAC uses adaptive probability modeling to achieve **excellent compression but is complex and slower**
- **StaticBAC eliminates runtime probability adaptation.**
  - Trades compression loss for speed and lower memory usage
- **Selects fixed probabilities from lookup tables**
- Processes tensors in **chunks**

Binary arithmetic coding steps:



What is new in StaticBAC V2?

V1:

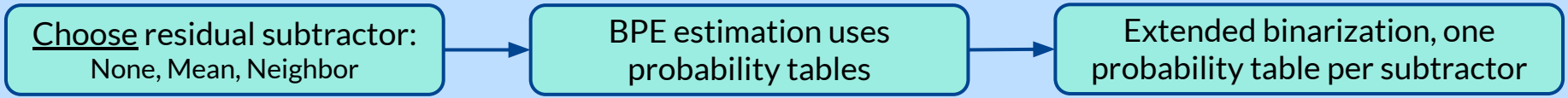


Always!

Decides Skip. Pessimist!

Too short unary sequence → Large remainder!
Probabilities based on DeepCABAC

V2:



Encoder chooses the best based on the BPE

More accurate estimation for Skip

Longer unary sequence → Shorter remainder
Updated probability tables wit V1 statistics

BPE: Bits per element

Evaluation methodology

Quantization determines the parameter distributions, so it must be evaluated together with AC.

Comparison uses three quantization methods:

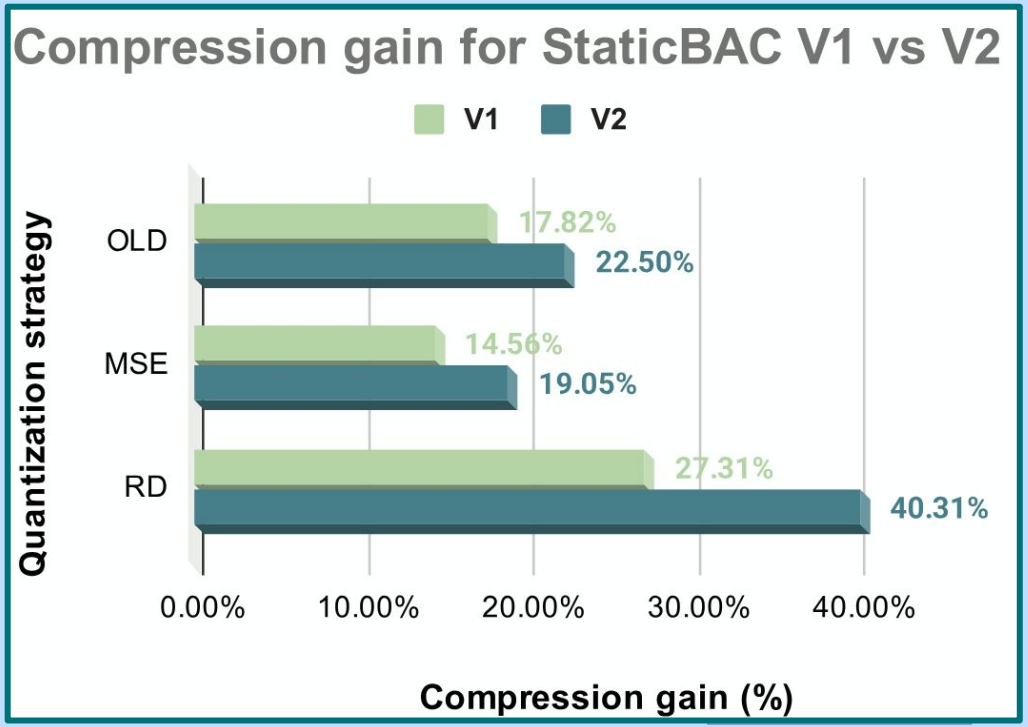
- **OLD:** V1's mixed-width QuantLayer + uniform MSE quantizer -- INT8 for most weights, INT12-16 all else
- **MSE:** uniform MSE quantizer -- INT8 for all types, least reconstruction error
- **RD:** rate-distortion (RD) quantizer, λ=0.15 -- INT8 for all types , reduces entropy

Evaluation on **13 neural network models:**

- CNNs and transformers
- Original models in FP32

Goal: Study how quantization affects compression for the two versions

Results - Overall compression comparison



Compression gain = size reduction compared to quantized input

V2's gains depend on the quantizer.

Compression improvement vs V1:

- +4.5% for MSE
- +13% for RD

V2 cost: **slower encoding.** Distribution dependent.

- MSE : encoder 66% slower and decoder 40% slower
- RD : encoder 21.4% slower and decoder 10% faster

Conclusion



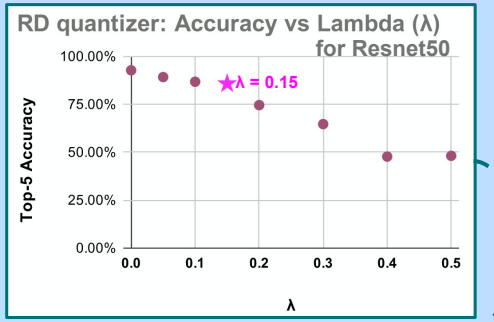
StaticBAC project

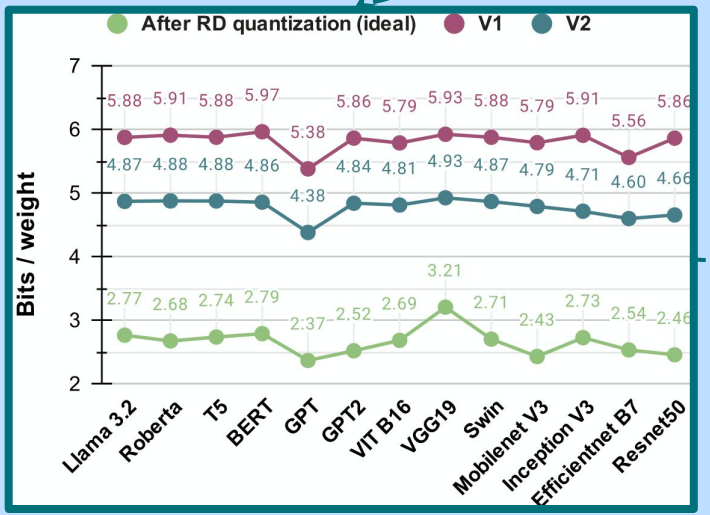
KEY TAKEWAY: **better compression depends on the symbol statistics and comes with trade-offs.**

- **40% average compression gain** using RD
- **RD** quantization creates **parameter distributions** that **V2** can **exploit more effectively**

For more results and implementation details please check the poster in the afternoon!

Results - BPE analysis and per model gains





StaticBAC V2 Compression Gain (%) for 3 quantizers			
Model	**OLD**	**MSE**	**RD λ 0.15**
ViT-B/16	36.92	18.4	39.83
EfficientNetB7	21.87	21.96	42.5
ResNet50	46.7	24.72	41.78
GPT	17.3	31.86	45.23
BERT	17.47	17.53	39.28
SwinV2-B	15.33	17	39.15
VGG-19	7.19	7.19	38.42
MobileNet-V3	16.91	17	40.12
Inception-V3	22.42	22.44	41.07
T5	18.89	18.89	39.04
roBERTa	14.15	15.31	39.03
GPT-2	41.71	19.62	39.48
Llama 3.2 - 1B	15.67	15.67	39.11

Important: the compression gains for RD consider a fixed λ=0.15 for experimental consistency. The choice of λ must be made individually, see:

TABLE XI: Inference evaluation for StaticBAC V2

Model	Inference task	Metric	Baseline	MSE	RD 0.075	RD 0.15
Resnet50	Image classification	Top 5 accuracy ↑	92.87%	92.78%	87.34%	85.80%
Eff.Net B7			96.90%	96.80%	86.97%	35.14%
VIT B16			95.32%	95.25%	95.07%	94.83%
BERT	Mask. lang. modeling	MLM loss ↓ / accuracy ↑	2.14 / 61.92%	2.19 / 61.06%	2.25 / 58.80%	2.34 / 58.35%
GPT2	Lang. model.	Perplexity ↓	29.56	56.31	140.68	235.3

Binarization flow comparison

